CYBERSECURITY LOG ANALYSIS REPORT

Student Answer Template

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Activity: CTM-ACTIVITY-8

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Instructions: Analyze each CSV dataset provided in the GitHub repository. Identify the likely attack type, cite specific log evidence, assess whether the attack succeeded, and explain the possible impact if the attack were successful. Do not rely on filenames or guesses; support your conclusions with the log entries.

ATTACK 1 ANALYSIS

1. Identification of Attack

- SSH Brute Force Attack.

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: Event ID BF-0001 / 2026-10-31T08:14:02Z
Observed activity: An SSH login attempt using the admin account failed.
Why it matters: It shows an attempt to access the system without the correct password.

Evidence 2: Event ID / Timestamp: Event ID BF-0003 / 2026-10-31T08:14:12Z
Observed activity: Multiple usernames were attempted from the same IP address.
Why it matters: This suggests someone is trying different accounts to gain access.

Evidence 3: Event ID / Timestamp: Event ID BF-0005 / 2026-10-31T08:14:22Z
Observed activity: The admin account was temporarily locked after repeated failures.
Why it matters: The system detected and blocked repeated login attempts.

3. Attack Analysis

- The attacker repeatedly tried to access the server using different usernames and passwords. The system detected the suspicious activity.

4. Attack Status

- The repeated login attempts were mostly blocked. However, a successful login using the student account from a different IP address was recorded. This needs further investigation to determine whether it was authorized.

5. Possible Scenario if the Attack Succeeds

- The attacker could gain unauthorized access, steal important information, or damage the system.

6. Prevention and Mitigation

Recommendation 1: Use strong and unique passwords.

Recommendation 2: Enable MFA (multi-factor authentication).

Recommendation 3: Use account lockout and login rate-limiting.

ATTACK 2 ANALYSIS

1. Identification of Attack

- Phishing Attack.

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: Event ID PH-0001 / 2026-10-31T09:02:11Z
Observed activity: An urgent email asked the employee to verify their account.
Why it matters: This is a common tactic used in phishing attacks.

Evidence 2: Event ID / Timestamp: Event ID PH-0003 / 2026-10-31T09:05:18Z
Observed activity: The employee clicked a suspicious link.
Why it matters: Clicking the link could expose the employee to a fake login page.

Evidence 3: Event ID / Timestamp: Event ID PH-0004 / 2026-10-31T09:05:41Z
Observed activity: The credential form was blocked by browser protection.
Why it matters: The security system prevented access to the suspicious login page.

3. Attack Analysis

- The attacker sent a fake email pretending to be from IT support. The email tried to convince the employee to click a link and provide account credentials.

4. Attack Status

- The phishing attempt was detected and mostly blocked. Although the employee clicked the link, the suspicious page was blocked. There is no evidence that the account was compromised.

5. Possible Scenario if the Attack Succeeds

- The attacker could steal the employee's login credentials and use them to access confidential company information.

6. Prevention and Mitigation

Recommendation 1: Avoid clicking suspicious links.

Recommendation 2: Check the sender's email address before responding.

Recommendation 3: Provide cybersecurity and phishing awareness training.

ATTACK 3 ANALYSIS

1. Identification of Attack

- Denial-of-Service (DoS) Attack, possibly a Distributed Denial-of-Service (DDoS) attack.

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: Event ID DOS-0002 / 2026-10-31T10:20:10Z
Observed activity: Traffic increased to 420 requests per second.
Why it matters: The traffic was much higher than the normal 35 requests per second.

Evidence 2: Event ID / Timestamp: Event ID DOS-0004 / 2026-10-31T10:20:30Z
Observed activity: Service latency increased, and some requests returned errors.
Why it matters: It shows that the server was having difficulty handling the traffic.

Evidence 3: Event ID / Timestamp: Event ID DOS-0006 / 2026-10-31T10:20:50Z
Observed activity: Traffic filtering and rate-limiting were enabled.
Why it matters: The system applied security measures to reduce the attack.

3. Attack Analysis

- The attacker sent a large amount of traffic from multiple IP addresses to the server. This caused service delays and errors, affecting normal users.

4. Attack Status

- The attack caused service disruption but was eventually mitigated. Traffic decreased, and the service returned to normal.

5. Possible Scenario if the Attack Succeeds

- The website or server could become unavailable, preventing users from accessing important services.

6. Prevention and Mitigation

Recommendation 1: Use DDoS protection services.

Recommendation 2: Apply traffic filtering and rate-limiting.

Recommendation 3: Monitor network traffic regularly.

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

- I think the DDoS attack could cause the greatest damage because it can make important websites and services unavailable. This can affect employees, customers, and the organization's operations. It may also cause financial losses.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

- Analyzing system logs is important because it helps identify suspicious activities, detect possible attacks, and understand what happened in a system. It also helps organizations respond quickly and prevent further damage.

3. What is the most important cybersecurity lesson you learned from this activity?

- The most important lesson I learned is that cybersecurity threats can happen anytime. Organizations should monitor their systems, use strong security measures, and educate employees to prevent cyberattacks.